

technological transfer and open science and will therefore assist in the maximisation of project outcomes and impacts. There will be no limitation placed on dissemination, exploitation, or communication of study findings.

### 2.3. *The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts*

As per this project's primary objective, findings will have enormous potential for scientific, societal, and economic impact because of their relevance to CD and the 40+ disorders/symptoms already localised by LNM<sup>13</sup>.

Brain disorders are the leading cause of disability and the second leading cause of death worldwide (approximately 144 per 100,000 people)<sup>1</sup>. rTMS has already demonstrated the ability to significantly reduce this burden, with 30-50% of patients with treatment-resistant major depression (who have already *failed prior antidepressant medication*) achieving remission with rTMS<sup>65</sup>. This has resulted in an estimated increase of 1.25 quality adjusted life years, superior to antidepressant medications (1.18)<sup>66</sup>. Similar response rates are seen in obsessive-compulsive disorder (~38%)<sup>67</sup>, migraine with aura (~39%)<sup>68</sup>, and smoking cessation (~33-44%)<sup>69</sup>. Therefore, it is extremely plausible that the present study could lead to the approval of rTMS protocols for clinical use in CD and other neurological disorders, resulting in similar reductions in the burden of disease; it is simply a matter of finding the correct targets and parameters for each population.

This study's findings also have the potential to improve the efficiency of healthcare delivery, significantly decreasing the associated economic costs. The total European cost of brain disorders was approximately €798 billion in 2010, with the average healthcare costs per person approximately €3,000 for major depression, €11,000 for Parkinson's disease, and €16,500 for dementia<sup>66</sup>. A treatment course of rTMS costs approximately €6,900, less than antidepressant medications (~€7,108), with cost-benefit analysis demonstrating the superiority of rTMS treatment<sup>70</sup>. rTMS does not require in-patient care, and comes without the economic costs associated with side-effects of medications, such as drug dependence<sup>66</sup>. New rTMS treatment protocols would also stimulate equipment manufacturing, technological advances in machinery, new magnetic pulse sequences to improve response rates, and the integration of MRI technologies to improve targeting of precise brain regions of interest<sup>22,23,38</sup>.

In summary, the short-term impacts of this project will mainly occur through publication of valuable new knowledge. In the medium-term, this knowledge will be leveraged to fund subsequent clinical trials in CD and other brain disorders. In the long-term, impacts will be demonstrated through the continued translation of these findings to unlock federally approved treatment protocols for a range of neurological and psychiatric disorders.

##SCOM-DIS-VIS-CDV## ##SIMP-ACT-IA##

## 3. **Quality and Efficiency of the Implementation** ##@QUA-LIT-QL@# ##@WRK- PLA-WP@# ##@CON-SOR-CS@# ##@PRJ-MGT-PM@#

### 3.1 *Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages*

#### 3.1.1 Work plan

The work plan is divided into five work packages and 23 tasks, the progress of which will be monitored by ten milestones and seven deliverables (Table 1). The plan has been designed after extensive consultation with supervisors Prof. Di Lazzaro and Prof. Joutsa, with consideration of the time it takes to complete administrative duties, train personnel, and plan and implement project methods, among other tasks. We each affirm the feasibility of the project based on our ability to complete prior TMS and fMRI projects of equal size and scope (e.g.<sup>16,41,43,44,59,71</sup>), and the extremely promising findings of our pilot study (Figure 3).

#### Work Package 1: Project planning, setup, and management

**Tasks:** Task **(T)1**. Planning of final study methods and statistical analyses; **T2**. Equipment inventory, setup, and testing; **T3**. Liaise with study personnel (e.g. radiographers to ensure fMRI data collection quality); **T4**. Patient recruitment and communication (explain study, schedule testing sessions, coordinate transport, directions, parking, reimbursement, obtain informed consent).

**Milestones:** Milestone **(M)1**. Ethics approval granted by FPUCBM ethics board

**Deliverables:** Deliverable **(D)1**. Mobility declaration; **D2**. Pre-registration of clinical trial; **D3**. Evaluation questionnaire and follow-up questionnaire.

#### Work Package 2: Training and transfer of knowledge

**Tasks:** **T1**. Meetings/rounds for clinical and medical training; **T2**. Develop key transferable skills (e.g. computer programming); **T3**. Train FPUCBM lab members in brain stimulation; **T4**. Set up LNM pipeline; **T5**. co-write grant with Prof. Di Lazzaro; **T6**. Train/complete fMRI data analysis; **T7**. Train Turku lab in brain stimulation.

**Milestones:** **M1**. Co-write grant with Prof. Di Lazzaro; **M2**. Start of secondment (fMRI data processing and analysis); **M3**. End of secondment (fMRI data analysed).

**Deliverables: D1. Career development plan.**

Work Package 3: Dissemination, exploitation, and communication activities

**Tasks:** T1. Communicate with dystonia foundations/patient groups; T2. Disseminate results at International Brain Stimulation Conference; T3. Disseminate results at International Congress of Movement Disorders; T4. Begin drafting publications; T5. Communicate outcomes on social and traditional media; T6. Exploit study findings in future grant applications; T7. Exploit findings to guide the methods of future clinical trials.

**Milestones:** M1. Begin drafting publications for pre-prints and peer-reviewed publications.

**Deliverables: D1. Plan for the dissemination and exploitation of results.**

Work Package 4: Data collection

**Tasks:** T1. Pilot data collection (testing two healthy participants to train collaborators, ensure correct methods and data quality); T2. Execute clinical trial (deliver real and sham rTMS to 30 patients with CD).

**Milestones:** M1. Pilot testing complete; M2. First clinical trial session; M3. End of last clinical trial session.

**Deliverables: D1. Safety and tolerability report (possible adverse events or patient discomforts during trial).**

Work Package 5: Data management and analysis (of clinical, behavioural, and brain stimulation data)

**Tasks:** T1. Create data analysis infrastructure (organise files, create systems and pipelines for processing of raw data for statistical analysis); T2. Analyse data in statistical software; T3. Upload data to open science repositories.

**Milestones:** M1. Data infrastructure created; M2. Clinical, behavioural, and brain stimulation data analysed (fMRI data to be analysed on secondment – WP 2).

**Deliverables: D1. Data management plan.**

| <b>Table 1. Gantt chart of the work plan, work packages (WP), milestones (M), and deliverables (D).</b> |        |     |    |    |    |    |   |    |   |    |    |    |        |    |    |    |    |    |    |    |     |       |
|---------------------------------------------------------------------------------------------------------|--------|-----|----|----|----|----|---|----|---|----|----|----|--------|----|----|----|----|----|----|----|-----|-------|
|                                                                                                         | Year 1 |     |    |    |    |    |   |    |   |    |    |    | Year 2 |    |    |    |    |    |    |    |     |       |
|                                                                                                         | 1      | 2   | 3  | 4  | 5  | 6  | 7 | 8  | 9 | 10 | 11 | 12 | 13     | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21  | 22    |
| <b>WP1</b>                                                                                              | ↓ ↓ ↓  | 80% | ↓  |    |    |    |   |    |   |    |    |    |        |    |    |    |    |    |    |    |     |       |
|                                                                                                         | D1     | D2  | M1 |    |    |    |   |    |   |    |    |    |        |    |    |    |    |    |    |    |     | D3    |
| <b>WP2</b>                                                                                              |        | ↓   | ↓  | ↓  | ↓  | ↓  | ↓ | ↓  |   |    |    |    |        |    |    |    |    |    | ↓  | ↓  | 60% |       |
|                                                                                                         |        |     |    | D1 |    |    |   | M1 |   |    |    |    |        |    |    |    |    |    | M2 |    |     | M3    |
| <b>WP3</b>                                                                                              |        |     |    | ↓  |    |    |   |    |   |    |    |    | ↓      |    |    |    |    | ↓  |    | ↓  | 30% | ↓ ↓ ↓ |
|                                                                                                         |        |     |    |    |    |    |   |    |   |    |    |    |        |    |    |    |    |    | D1 | M1 |     |       |
| <b>WP4</b>                                                                                              |        |     |    | ↓  | ↓  |    |   |    |   |    |    |    |        |    |    |    |    |    |    |    |     |       |
|                                                                                                         |        |     |    | M1 | M2 |    |   |    |   |    |    |    |        |    |    |    |    |    | M3 | D1 |     |       |
| <b>WP5</b>                                                                                              |        |     |    | ↓  |    |    |   |    |   |    |    |    |        |    |    |    |    |    | ↓  |    | 5%  | ↓     |
|                                                                                                         |        |     |    | M1 |    | D1 |   |    |   |    |    |    |        |    |    |    |    |    | M2 |    |     |       |

**Note:** WP1: Project planning, setup, and management; WP2: Training and transfer of knowledge; WP3: Dissemination, exploitation, and communication activities; WP4: Data collection; WP5: Data management and analysis. Red arrows ↓ show timing of each task within each WP. Vertical line after month 18 indicates secondment start. Percentage values correspond to % of total workload allocated to WP at that stage of fellowship.

### 3.1.2 Risk mitigation strategies

**Risk:** Delays in ethics being granted by FPUCBM ethics board leading to later start date

**Impact/Probability:** medium/low

**Mitigation strategies:** Many similar projects from Prof. Di Lazzaro's lab have been granted ethics approval by this review board in approximately two months. To mitigate this risk, I will initiate the application prior to my arrival in Italy so that we can identify potential obstacles ahead of time.

**Risk:** Difficulty in recruiting CD patients leading to inability to test 30 patients in 18 months

**Impact/Probability:** high/low

**Mitigation strategies:** We will recruit patients from Prof. Di Lazzaro and the 10+ neurologists within the Unit of Neurology, who each see approximately five CD patients per month. If recruitment is slower than expected, we will also recruit from neurologists at other hospitals in the region, such as from collaborator Prof. Koch at Foundation Santa Lucia (section 1.3.1).

**Risk:** Poor quality data

**Impact/Probability:** high/low

**Mitigation strategies:** I have already delivered the CD clinical scale for a previous project (section 1.4). To ensure quality, I will undertake further training by FPUCBM neurologists prior to data collection and confirm high inter-rater reliability. If kinematic device and machine learning outcomes (section 1.2.1) do not correlate with clinical scale outcomes, we will perform analyses to investigate why. Prof. Joutsa and I have already planned the specific fMRI parameters (section 1.2.1). However, we will review these again to ensure they align with the latest methods.